meters (one hundred billion trillion-trillionths of a meter) across. The German physicist Max Planck, after whom these unimaginably small quantities are named, introduced the idea of quantized energy in 1900 and is generally credited as the father of quantum mechanics.

The clash between gravity and quantum mechanics poses no practical problem for the contemporary universe. Astrophysicists apply the tenets and tools of general relativity and quantum mechanics to very different classes of problems. But in the beginning, during the Planck era, the large was small, and we suspect there must have been a kind of shotgun wedding between the two. Alas, the vows exchanged during that ceremony continue to elude us, and so no (known) laws of physics describe with any confidence the behavior of the universe over that time.

We nonetheless expect that by the end of the Planck era, gravity wriggled loose from the other, still unified forces of nature, achieving an independent identity nicely described by our current theories. As the universe aged through 10^{-35} seconds it continued to expand, diluting all concentrations of energy, and what remained of the unified forces split into the "electroweak" and the "strong nuclear" forces. Later still, the electroweak force split into the electromagnetic and the "weak nuclear" forces, laying bare the four distinct forces we have come to know and love: with the weak force controlling radioactive decay, the strong force binding the atomic nucleus, the electromagnetic force binding molecules, and gravity binding bulk matter.

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A trillionth of a second has passed since the beginning.

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All the while, the interplay of matter in the form of subatomic particles, and energy in the form of photons (massless vessels of light energy that are as much waves as they are particles) was incessant. The universe was hot enough for these photons to spontaneously convert their energy into matter-antimatter particle pairs, which immediately thereafter annihilate, returning their energy back to photons. Yes, antimatter is real. And we discovered it, not science fiction writers. These transmogrifications are entirely prescribed by Einstein's most famous equation: $E = mc^2$, which is a two-way recipe for how much matter your energy is worth, and how much energy your matter is worth.